

TARAZ, Kazakhstan

WMO: 383410

Lat: 42.850N

Lon: 71.300E

Elev: 2139

StdP: 13.59

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	1.4	-9.2	3.6	-4.9	-2.6	5.2	2.3	35.2	42.6	27.3	41.6	3.4	190	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	97.1	63.8	94.6	63.5	91.7	62.7	67.2	87.8	66.0	87.0	64.8	85.3	8.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	87.0	71.8	59.2	81.4	71.9	57.5	76.4	72.3	32.9	87.6	31.9	87.1	31.0	85.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	19.7	15.2	-10.0	102.7	9.3	2.8	-16.7	104.8	-22.1	106.4	-27.3	108.0	-34.0	110.1		
			-9.9	70.0	9.0	2.2	-16.4	71.5	-21.6	72.8	-26.7	74.0	-33.3	75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	27.0	29.8	42.9	54.4	64.3	73.7	77.2	74.7	64.6	52.7	39.5	29.8
	DBStd	19.99	11.83	12.16	10.50	8.54	6.64	5.92	5.03	5.75	7.62	9.11	11.31	11.97
	HDD50	2649	714	568	263	53	1	0	0	2	78	342	628	
	HDD65	5629	1178	985	685	329	94	7	1	5	99	390	764	1092
	CDD50	3627	1	3	44	184	444	710	843	767	440	162	28	1
	CDD65	1132	0	0	1	10	72	267	379	307	87	9	0	0
	CDH74	14433	0	0	16	172	918	3341	4852	3871	1107	147	9	0
	CDH80	6774	0	0	2	36	281	1571	2516	1918	410	40	0	0
(14) Wind	WSAvg	6.1	5.3	5.4	6.3	6.8	7.0	6.7	6.5	6.8	6.0	5.2	5.1	5.5
(15) Precipitation	PrecAvg	13.1	1.1	1.2	1.5	1.9	1.6	1.0	0.4	0.2	0.4	1.1	1.5	1.3
	PrecMax	23.5	5.5	4.2	4.4	4.8	4.7	5.8	2.0	1.8	2.3	2.8	4.5	3.3
	PrecMin	6.8	0.1	0.2	0.4	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.7	0.9	0.7	0.9	1.1	1.1	1.2	0.5	0.4	0.5	0.7	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	63.1	75.5	84.5	91.1	98.9	102.2	100.5	93.6	85.7	73.9	59.9
		MCWB	45.4	47.7	53.6	60.5	61.8	64.2	64.3	64.1	61.8	58.0	54.0	45.4
	2%	DB	51.2	55.5	69.4	79.1	86.3	95.1	97.7	96.3	88.8	77.4	65.8	53.6
		MCWB	41.8	44.1	52.3	58.8	61.4	63.3	64.5	63.3	60.1	55.5	50.4	43.4
	5%	DB	45.7	49.8	64.2	74.0	82.8	91.9	94.7	93.3	84.6	72.9	60.2	48.5
		MCWB	39.0	41.7	50.2	57.4	60.9	62.8	63.7	63.0	58.9	53.2	48.2	40.7
	10%	DB	41.2	44.8	59.0	69.6	79.1	89.2	91.7	90.1	80.6	68.1	55.3	44.4
		MCWB	36.3	39.2	48.8	55.6	59.9	62.2	63.2	62.2	57.5	51.6	46.0	38.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	49.3	56.4	64.2	67.0	68.8	69.1	68.8	63.9	58.6	55.7	46.9
		MCDB	56.1	60.9	70.0	77.3	82.2	89.9	89.9	90.2	87.4	79.4	67.0	55.6
	2%	WB	42.5	45.8	53.4	61.0	64.7	66.4	67.3	66.2	61.7	56.5	52.2	44.3
		MCDB	49.9	54.0	65.1	73.6	80.4	86.8	88.1	89.6	84.4	73.9	63.5	52.0
	5%	WB	39.3	42.2	51.2	58.9	62.9	65.2	66.2	64.8	59.9	54.7	49.2	41.4
		MCDB	44.8	48.6	62.0	71.2	77.9	84.7	87.8	87.2	81.6	69.4	59.2	47.5
	10%	WB	36.6	39.3	49.1	56.7	61.3	63.9	65.0	63.5	58.1	52.8	46.4	38.6
		MCDB	40.8	44.8	58.3	67.6	75.3	82.9	86.3	85.5	78.6	65.7	53.6	43.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.2	18.5	20.6	23.7	25.6	26.1	27.3	25.7	21.7	17.3	16.0
		MCDBR	22.1	22.9	26.9	28.2	28.9	29.7	29.7	30.1	29.6	28.4	25.0	22.8
	5% WB	MCWBR	14.4	13.9	13.7	13.6	11.8	10.0	9.7	10.4	11.8	13.0	13.4	14.2
		MCWBR	20.0	21.2	24.0	24.0	24.4	25.5	25.7	27.1	27.8	24.5	22.4	20.4
(40) Clear-Sky Solar Irradiance	taub		0.333	0.370	0.434	0.453	0.434	0.425	0.416	0.385	0.365	0.399	0.381	0.326
		taud	2.154	2.082	1.990	2.005	2.128	2.189	2.221	2.298	2.306	2.107	2.091	2.167
	Ebn at Noon		245	254	252	258	267	270	271	276	273	240	221	234
		Edh at Noon	31	40	50	53	48	46	44	39	36	39	34	28
(44) All-Sky Solar Radiation	RadAvg		591	861	1152	1557	2028	2299	2312	2132	1679	1032	601	476
	RadStd		61	99	96	180	141	118	122	95	85	124	82	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-2.05	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page